

2 Design Principles of Nanocarriers

Size, Shape, and Surface Properties, Stimuli-Responsive Systems

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2.1 INTRODUCTION

Nanocarriers are one of the most important developments in therapeutics' delivery systems of nucleic acids and proteins. Targeted drug-delivery systems should always consider the size and shape of the nanocarriers to have optimum size, shape, surface properties, and stimuli responsiveness. This will include fundamental aspects of the design of nanocarriers, enlightening their paramount role in increasing the therapeutic efficacy and specificity of drug delivery for other complex diseases, but primarily cancer.

The size of nanocarriers plays a pivotal role in their biodistribution, cellular uptake, and clearance from the body. Nanocarriers in the 10–200 nm size range can use the enhanced permeability and retention (EPR) effect to target the tumor efficiently. In this effect, the vasculature of the tumor tissues is enough to be considered leaky, allowing nanocarriers to accumulate passively [1]. In addition, the circulation half-life of the nanocarriers, which this size would strongly influence, might be subject to being taken up by the mononuclear phagocyte system (MPS) and undergoing renal clearance [2].

The shape of the nanocarrier highly determines how it will interact with the biological environment: cellular internalization and biodistribution. Namely, nanocarriers that are non-spherical in shape (rods or discs) preferably accumulate in the tumor tissue and thus may be more effective than spherical ones in treating the outcomes [3]. The aspect ratio and the surface curvature of the nanocarrier may further enhance their permeation through the biological barriers and help them evade immune recognition [4].

In addition, surface modification of the nanocarriers comprises targeting ligands or stealth coatings, such as polyethylene glycol (PEG), to increase the specificity of the nanocarrier and avert host defenses. This way, those nanocarriers with targeting ligands are fixed to specific cell types or tissues; hence, the delivery efficiency to the site of action is increased. Besides, the use of nanocarriers protects against opsonization by surface modification, which allows them to have a long time of exposure in circulation in the blood [5]. Furthermore, the use of nanocarriers allows protection against opsonization by surface modifications.

Thus, the stimuli-responsive nanocarriers become an excellent choice for delivering the therapeutic payload in response to a specific cue, such as pH, temperature, or enzymatic activity, that allows controlled drug release at the site of action. For example, this can be employed using the acidic tumor microenvironment or endosomal pH for drug release, thereby enhancing the therapeutic index while concomitantly diminishing systemic toxicity [6].

The design principles of nanocarriers are integral to their function as drug-delivery systems (Figure 2.1). In terms of size, shape, surface properties, and incorporation of stimuli-responsive elements, nanocarriers are becoming suitable for reaching the target delivery of nucleic acids and proteins in varying kinds of diseases. The remaining problems in the area include better specificity in targeting, reduced off-target effects, and increased stability of nanocarriers. Further development in the design of nanocarriers would probably bring the greatest revolution in therapeutic delivery, helping to realize effective and personalized medical treatments.

2.1.1 Size

The size of nanocarriers is a crucial parameter in their design; it dramatically influences the biodistribution, cellular uptake, and clearance of nanocarriers and, eventually, will affect the therapeutic efficacy of drug-delivery systems. Understanding the optimal size ranges of nanocarriers is thus important in rational design because it will enable these systems to pass effectively through the complex biological environment to the targeted sites.

The biodistribution of nanocarriers is heavily dependent on their size. For example, most tumors have been shown to be highly leaky [2]; therefore, nanoparticles (NPs) in the range of 10–100 nm preferentially accumulate in tumor tissues through the effect of EPR. This size-dependent accumulation can significantly impact the therapeutic efficacy of anticancer drugs. Otherwise, when the size of NPs exceeds 200 nm, the MPS is more likely to recognize and remove them. This reduces their time of circulation in the body, and they may thus not reach the site of action [1].

In addition to the above, the nanocarrier's size also influences the cells' uptake. Internalization into the cells of particles below 50 nm in size might be much better than that of particles exceeding 50 nm through several endocytosis routes. For example, it has been reported that particles of 50 nm have shown higher uptake rates by cancer cells, thus increasing efficiency in drug delivery to target sites [7]. However, NPs of a

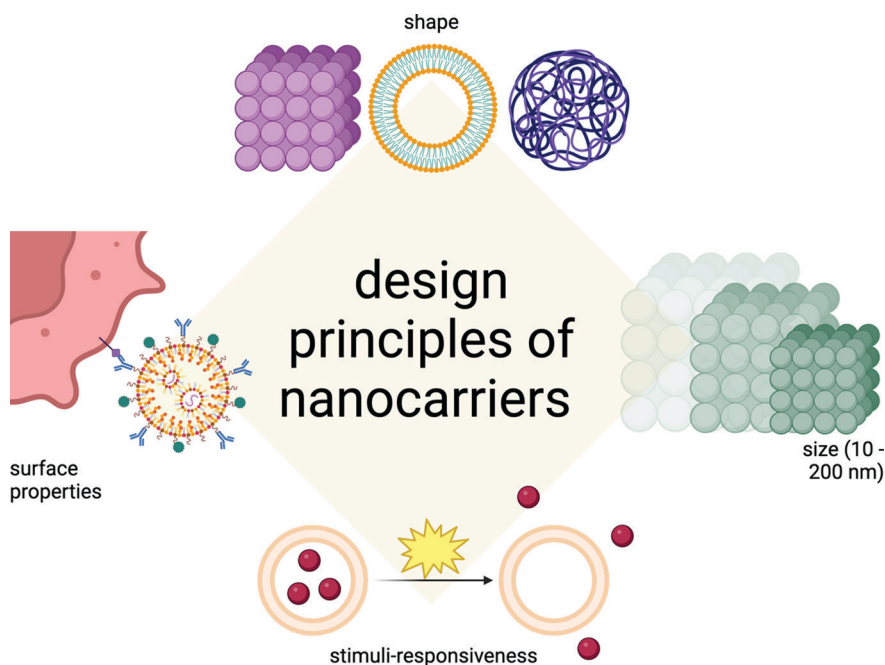


Figure 2.1 Design principles of nanocarriers. This schematic illustrates the key factors influencing nanocarrier design and function: shape (represented by spherical, liposomal, and polymeric structures), size (ranging from 10 to 200 nm), surface properties (depicted by a functionalized nanoparticle with various ligands), and stimuli responsiveness (shown as a nanocarrier releasing its payload upon stimulation). These design principles are crucial for optimizing nanocarrier performance in drug delivery, affecting biodistribution, cellular uptake, and therapeutic efficacy. The interplay between these factors determines the nanocarrier's ability to overcome biological barriers and achieve targeted drug delivery.

size lower than 10 nm can rapidly escape from the target tissue due to diffusion and subsequently be excreted via the kidneys, thus compromising the delivery of the therapeutic payload into the target cells [8].

The clearance of nanocarriers from the body is another crucial aspect influenced by their size. Particles smaller than 10 nm are quickly cleared from circulation by renal filtration, whereas those larger than 200 nm are predominantly cleared by the liver and spleen [9]. A general optimal size between 10 and 100 nm is always preferable to effectively penetrate tissues at a trade-off for the most effective circulation time without being cleared too early, thereby maximizing therapeutic efficacy.

Ideal size ranges for these are from 10 to 100 nm and are preferred for application and targeting tissue. In the case of tumor targeting through the EPR effect, sizes ranging around 10–100 nm of the nanocarriers are preferably used, allowing deep penetration into tumor tissue and efficient cell uptake. Those of a size toward the upper limits of this range might find the best balance between tissue accumulation and clearance in applications where passive targeting for long circulation times is required, for example, toward cardiovascular diseases or inflammatory sites [10].

Size is one of the most important criteria influencing nanocarriers' biodistribution, cellular uptake, and clearance, thereby affecting therapeutic success. Therefore, size tuning of the nanocarriers is crucial for optimizing their delivery efficiency with reduced off-target effects, which increases therapeutic efficacy. Future developments in the design of the nanocarrier will further delve into the complex interplay of NP size in biological interactions, essential for drug-delivery systems to be optimized against the broadest range of therapeutic applications.

2.1.2 Shape

The shape factor in nanocarrier design remarkably influences their biological interaction, including circulation time, cellular internalization, and targeting ability. Later research on other geometrical forms, such as spherical, rod-shaped, and many other geometries, demonstrated that the exhibited behaviors were geometrical form-dependent, which can be harnessed to optimize drug delivery and therapeutic outcomes.

The size, shape, and flexibility of nanocarriers significantly influence their blood circulation time. Historically, spherical NPs were preferred for their ease of synthesis and uniformity. Recent studies, however, show that elongated shapes like rods or worms demonstrate increased circulation times due to reduced uptake by the MPS compared to spherical NPs. This extended circulation enhances the nanocarrier's ability to reach and accumulate at the target site, improving therapeutic efficacy [11, 12].

Additionally, the nanocarrier's shape also affects the cell's internalization. Because of endocytosis, in a spherical-shaped NP, penetration into a cell would be evident without any doubt. For instance, rod-shaped or other elongated NPs were better internalized and followed several routes, including macropinocytosis or receptor-mediated endocytosis, depending on the aspect ratio and surface modifications [13]. This enhanced and selective uptake can be leveraged to improve drug delivery to specific cell types or tissues.

Geometry has been recognized as an important determinant of the targeting capability of nanocarriers. Spherical NPs target tumor sites passively through EPR, but rod-shaped and other nonspherical NPs have evidenced better-targeting abilities. Geometries of this kind may help better interact with some cell receptors and increase the binding affinity of target cells. For example, rod-shaped NPs can easily penetrate deeper into tumors and thus are more effective in producing improved therapeutics for cancer [14].

Further studies of several geometrical shapes in the case of NPs bring out quite interesting features and possible applications. NPs with a spherical shape are especially remarkable for facile syntheses and effective cellular uptake [15]. However, these shapes have shorter circulation times and a lower precision of targeting than other shapes [16]. On the other hand, rod-shaped NPs have the advantage of distribution over time and have been attributed to their elongated form. This property supports their presence not only in the bloodstream but also increases the penetration depth of the tumors, which allows better localization of the drug and may, in turn, improve therapeutic efficiency [17].

The exploration of NP designs extends beyond rod-shaped and spherical forms, incorporating discoidal, worm-like, or cubic shapes. Each shape offers unique opportunities for optimizing nanocarrier design, allowing for the precise tuning of formulations in combination with specific compounds. This variety in shape enables the development of distinct biodistribution profiles, targeting capabilities, and cellular uptake mechanisms, catering to specific therapeutic needs. Such diversity in NP design is crucial for enhancing the effectiveness of targeted drug-delivery systems, ultimately improving therapeutic outcomes [18, 19].

2.1.3 Surface Properties

All the surface features of nanocarriers, like their charge and changes that make them more soluble, stable, and specific to targets in the immune system and clearance, are due to their strong interaction with living systems. Therefore, these properties could be refined for better biocompatibility, targeting efficiency, and therapeutic efficacy toward effective drug-delivery applications.

It has been reported that the surface charge of nanocarriers considerably affects their interaction with biological membranes, leading to consequences for their biodistribution, cellular uptake, and clearance. Dey et al. [20] point out that the surface charge is the most crucial factor in modulating the physiology of the macrophage, and it has to be adequately optimized for gene delivery and mRNA vaccine delivery. Surface changes like PEGylation and ligand attachment were used to do this. These changes are meant to make the nanocarriers more stable and soluble while also making them more specific to their target.

On the other hand, Gajbhiye et al. [21] explained that PEGylation greatly extends the circulation time of the nanocarriers because it dramatically decreases their uptake by the Reticuloendothelial System (RES). They regarded it as a rather predominant and critical mechanism that results in upregulation of nanocarrier uptake by the brain. Furthermore, Milani et al. [22] go further to outline how the cellular uptake of targeted nanocarriers, along with their nanotoxicity and site specificity, has been upwardly modulated toward lower toxicity and higher site specificity as one strategy of general therapeutic enhancement.

The interaction of nanocarriers with the immune system is profoundly affected by their surface chemistry. Vincent et al. [23] further illustrate that surface chemistry-mediated modulation of adsorbed albumin folding state may be one of the mechanisms to guide the clearance of nanocarriers by the distinct *in vivo* macrophage subset, hence promoting evasion or nonspecific clearance. This finding is pivotal for designing nanocarriers to avoid premature clearance and achieve targeted delivery. Additionally, attaching antibodies to the nanocarrier surface first ensures they work and can target cells when the biomolecular corona is added later. This also suggests that the properties of the surface are conserved, and the functionality toward targeting the nanocarriers can even be provided in the presence of the biomolecular corona [24].

Nanocarriers' surface properties, such as charge, solubility, and stability, are fundamental in interacting with the immune system. These properties directly affect how well they target and treat diseases. Nanocarriers can function better for specific therapeutic uses by engineering and functionalizing their surfaces to fine-tune specific properties. This improves drug delivery and treatment effectiveness. This approach to surface engineering improves the delivery and function of nanocarriers and opens new avenues for overcoming the

challenges of biological systems for efficient, targeted drug delivery. As a result, there is a growing focus on surface modification techniques to develop nanocarriers with improved performance in clinical applications [23, 25].

2.2 INTRODUCTION TO STIMULI-RESPONSIVE NANOCARRIERS

Stimuli-responsive nanocarriers represent a new shift in targeted drug-delivery technology and stand at the frontier of such development. They offer the most sophisticated system using environmental cues to release therapeutic agents where they are most required. These are well-tuned systems to a vast number of internal and external stimuli, going from biochemical ones, such as pH and redox potential, to physical ones, like temperature, magnetic fields, and light (Figure 2.2). At this line, the response is the basis for high efficiency in treatments that will offer, at the same time, dramatically improved therapeutic outcomes and minimize the risk of unwanted systemic side effects.

The hallmark of these nanocarriers is their fine-tuned ability to deliver adroitly in a controlled mode in response to different physiological statuses or outside environments. This precise targeting mechanism has not made an enormous difference in ensuring that therapeutic agents stay where they are needed most and that only a small amount of drugs reach healthy tissues to avoid harmful side effects and get the most out of the therapy. For example, pH-sensitive nanocars can be designed in such a way that they can make use of the acidity difference between the normal tissue and the acidic microenvironment of the tumors. It thereby facilitates selective antitumor drug release at the tumor site. Efficiency is very high in this particular method of drug release, according to Sethuraman et al. [26].

Among the intrinsic bodily conditions, the nanocarriers are responsible for detecting are changes in pH, redox potential, and the concentration of specific enzymes, among others. These are the conditions at the pathological site, such as the acidic pH found in tumor environments and the high glutathione concentration in cancer cells, which may be used as a trigger point for the release of drugs. Handa et al. [27] thus have given compelling evidence on the effectiveness of utilizing these internal stimuli for targeted therapy. External stimuli include temperature, magnetic fields, light, and ultrasound, which underlie non-invasive therapeutic interventions, allowing conduction and possessing more precision of time. Guo and You [28] created a new type of light-responsive nanocarriers that are mostly excited by near-infrared (NIR) light. These nanocarriers have much potential for controlled drug release while penetrating deep into tissues with the least damage to healthy tissue around them. This is an auspicious direction for controlled drug delivery.

Targeted and controlled drug delivery by stimuli-responsive nanocarriers points to the utmost necessity in this area of several specialties, including cancer therapy. This means the combination of redox, light, and pH stimuli with cyclodextrin-based nanocarriers offers a versatile ground for the controlled release mechanism of the chemotherapy agent [29]. Meanwhile, dual stimuli of pH and redox conditions have attracted more attention, such as being highly demanded to a greater extent in advancing technology for

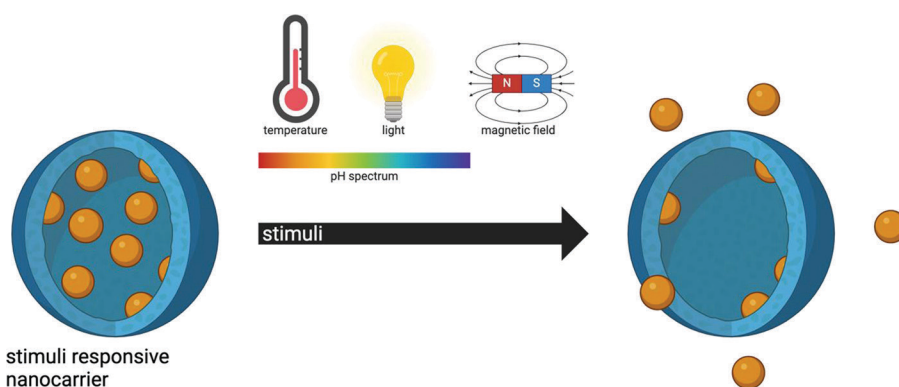


Figure 2.2 Stimuli-responsive nanocarriers and their activation mechanisms. This figure demonstrates the concept of stimuli-responsive drug release from nanocarriers. On the left, a loaded nanocarrier is shown before stimulation. Upon exposure to specific stimuli (represented by icons for temperature, light, magnetic field, and pH), the nanocarrier undergoes structural changes, leading to the release of its payload (orange spheres). The pH spectrum indicates the range of acidic to basic environments that can trigger response. This design allows for precise control over drug release, enhancing therapeutic efficiency and reducing off-target effects in applications such as cancer treatment.

nanocarrier development [30, 31]. This will ensure the delivery of the drugs to tumor sites more effectively and specifically, marking the advent of new, improved strategies for treating cancer [32, 33].

The continuous evolution and refinement of stimuli-responsive nanocarrier designs bear immense promise for the future of medical treatment, especially toward more exact, targeted therapeutic interventions. Technologies such as multi-stimuli-responsive biocompatible magnetic nanocarriers, responsive nanostructures [34], and stimulus-responsive antibacterial therapy [35] underscore the potential of such designs to revolutionize how diseases are treated. The development of stimuli-responsive image-guided nanocarriers [36] and nanoassemblies targeting tumors and their microenvironment [37] further illustrates the importance of these innovations in healthcare, highlighting the pivotal role of surface engineering and functionalization toward efficient targeted drug delivery.

2.2.1 Design and Application of pH-Responsive Systems

The pH-responsive nanocarriers target all the unique acidic conditions prevailing in the complex and heterogeneous tumor microenvironment, as well as those present across cellular compartments, including the endosome and lysosome, to optimize the site-specific therapeutic agent delivery. Thus, these advanced systems become crucial for increasing drug-delivery selectivity and efficiency, and these applications turn out to be very promising, mainly in the scope of cancer treatment.

Thus, pH-responsive nanocarriers may act through a different mechanism under acidic conditions, including protonation, charge inversion, or breaking chemical bonds. Those mechanisms could result in the conformational changes of nanocarriers, and because of that, they are able to deliver the drug with precision to the places of their action. For example, the acid microenvironment, typical in tumor tissues, may trigger functional groups' protonation at the nanocarrier's surface. As a consequence of protonation, these groups can reverse the charge and thereby increase the uptake of nanocarriers by cells or directly release the encapsulated drug in the targeted tissue or cellular compartment [38]. The strategic design of such pH-responsive systems ensures therapeutic agent delivery in a controllable way right at the target site, minimizing side effects while maintaining maximal therapeutic efficacy.

For instance, pH-responsive magnetic nanocarriers designed with charge reversal capability can guide subcellular doxorubicin delivery for chemotherapeutic drug delivery to the target, which is the acidic environment of the tumor [38], and also to multi-stimuli-responsive behaviors. This kind of micellar nanoassembly, therefore, is responsive to pH and acid-labile group incorporation into the hydrophobic cores, which presents a very promising possibility for the target release of drugs in a sequence triggered by acidic pH found within a tumor site or intracellular compartments [39]. The principle of pH responsiveness is of the essence when developing nano self-assemblies of anticancer drugs forming structured carriers, which can disassemble under an acidic environment for therapeutic payload release [40]. The second major advantage is the more generalized application of the pH-responsive micro- and nanocarrier system strategy because it offers a versatile platform for delivering a more significant number of therapeutic agents [41]. Lastly, modern design concepts for advanced micelles include charge-conversional and pH-sensitive polymeric micelles, ensuring more effective intracellular drug delivery to facilitate better drug therapeutic efficacy [42].

A recent study by Kanamala et al. [43] has shown that nanocarriers are an effective pH-sensitive drug vehicle for targeting tumors. They conclude that such nanocarriers overcome the drawbacks of conventional chemotherapy, making it possible for improved therapeutic outcomes, as they are formulated with pH-sensitive biomaterials to allow drug release from the carrier system at the tumor site, both extracellularly and intracellularly. Similarly, Nogueira-Librelotto et al. [44] furthered the study of pH-responsive chitosan–tripolyphosphate NPs functionalized with a pH-sensitive amphiphile for increased intracellular uptake and the resultant antitumor activity of doxorubicin. This underlines the critical, innovative function of pH-responsive systems in cancer therapy.

This is a very active area of research, with recent progress targeting drug-delivery systems that are acid-sensitive when exposed to the pH conditions of the tumor microenvironment or even when exposed to the usual pH values in endosomes and lysosomes. The pH-responsive nanocarriers developed by Wu et al. [45] can change their programmed size, and these carriers are supposed to enhance the safety and therapeutic efficacy of antitumor drugs by showing adaptability to pH changes. In further research, Prajapati et al. [40] designed pH-responsive nano self-assemblies from 2-hydroxyoleic acid and explored them as a novel platform for the targeted delivery of chemotherapeutics to cancer cells to allow localization in cancer cells and minimize toxic side effects.

All these examples have shown that leading strategies are being used in relation to pH-responsive nanocarriers in the field of targeted drug delivery for cancer therapy. One of the main advantages of pH-responsive nanocarriers is the possibility of achieving either selectivity toward targets or reactivity in the acidic environment characteristic of the tumor or toward the pH values existing in the cell. This allows controlled drug release, together with localized drug targeting. This way, systemic toxicity is significantly

reduced, and at the same time, improved patient care outcomes are observed [46]. The versatility of cyclodextrin-based nanocarriers may find applicability in the encapsulation of redox, light, and pH stimuli, thus depicting the same. Controlled release mechanisms of the respective chemotherapeutic agents and bringing new inventive strategies in this area of cancer therapy [47]. Moreover, dual stimuli, such as pH and redox conditions, have gained huge research attention and are forthcoming due to the demands of nanocarrier technology. Such developments further assure more effective and specific delivery of drugs into neoplastic sites, heralding the first glimmers of novel, improved strategies in the treatment of cancer [48].

pH-responsive nanocarriers have allowed new scopes in the stimuli-responsive drug-delivery landscape. Such nanocarriers are designed in such a way that they can release drugs at a controlled rate, exactly at the site of the tumor, using the specific acidic microenvironment of the tumor. This specificity becomes possible due to the unique composition of tumor tissues, unlike healthy cells. This means the respective pH-sensitive nanocarriers could target cancer cells with much-improved, enhanced accuracy, significantly improving the ability of anticancer treatments carried out inside cancer cells and, at the same time, apparently reducing risks associated with systemic toxicity [46]. These pH-responsive nanocarriers create three main processes: protonation, charge inversion, and the cleavage of chemical bounds. These processes cause changes in conformation that make it easier for therapeutic agents to get to the right place where they need to go. For instance, the tumor's acidic environment protonates the nanocarrier's functional groups, inducing charge reversal, which is characteristic of promoting cellular uptake, or drugs are released directly in the respective tissue [48].

Constantly, very active research and development in this area are expanding the horizons of personalized medicine and precision medicine and opening new pathways for the treatment of many diseases. In leading innovation, pH-responsive systems promise to improve the healthcare outcome of a product that offers safety and effective treatment tailored to the unique needs of an individual patient [49].

2.2.2 Temperature and Light-Responsive Systems

Within just a couple of years, advanced smart nanocarriers have become increasingly of central importance in the research of targeted drug delivery with the development of nanotechnology. With that in place, smart nanocarriers are tuned in such a way that they are sensitive to certain stimuli that include temperature and light, therefore guiding and governing the site-specific release of therapeutic agents within the tumor site to increase precision and effect of treatment, with collateral damage to healthy tissues minimized [50]. For example, the slight temperature change we discussed earlier may cause the temperature-responsive nanocarriers to change how permeable they are. This will ensure that drugs are released at temperatures between the healthy tissue and the tumor site. The constituents of this unique characteristic are those materials that can physically or chemically change at given temperatures, which will permit the drug to be released in a targeted and controlled way [51].

Light-responsive nanocarriers work in a similar way to spatial drug release. In this case, light-responsive nanocarrier activation might occur only within a tumor; thus, therapeutic agents will be released only at the place of interest. It not only increases drug concentration at the tumor site for increased therapeutic effect but also, in a way, significantly reduces the systemic circulation of highly potent drugs, which thereby reduces toxicity risk [52]. A temperature- and light-responsive unit was thus implemented into a single nanocarrier platform for synergistic effects toward efficient anticancer phototherapy. The nanocarriers with two functions can use light and high temperatures to break down drugs more efficiently in photothermal molecules, improving cancer cell treatment [52].

This progressive improvement and application of these stimuli-responsive nanocarriers underline these technologies' potential usefulness in revolutionizing cancer treatment approaches. Greater development in this area of research holds promise toward expanding the tenets of personalized and precision medicine, thereby opening new opportunities to improve health outcomes within the healthcare setting via targeted treatment of cancer and other diseases [53].

In the field of drug-delivery systems, there has been a significant advancement in the form of temperature-sensitive liposomes and NPs. These systems are designed to launch the release of the drug at specific temperatures, which is a significant development. This is a targeted approach and can, therefore, be of great value in cancer therapeutics, where it is indispensable to deliver therapeutic agents specifically into the tumor sites while leaving the rest of the system with minimized side effects. Choi et al. [54] first set off the development of radially porous structured silica NPs to demonstrate their potential to load large amounts of drugs for temperature-controlled release. On such an issue, it was found that the synthesized NPs have immense potential for the controlled release of ibuprofen, one of the most common anti-inflammatory and analgesic drugs. This innovation presented an application of practical value with regard to temperature-sensitive nanocarriers and improved drug-delivery efficiency.

Pandey et al. [55] also looked into the behavior of temperature-responsive fluorescent polymer nanoparticles (TRFNPs) to learn more about the temperature-responsive system. The designed NPs not only have the ability to act as effective cellular imaging agents but also indicate the controlled release of drugs into the living cells of an anticancer drug. Thus, they point out another vast potential for applying TRFNPs in the domains of diagnostics and therapeutics in nanomedicine. Bearing these bases in mind, recent research has followed up on different applications and novelties brought by temperature-sensitive nanocarriers. Lohe and Umekar [56] have provided thorough and elaborate review articles on the thermosensitive liposomal drug-delivery system, focusing on the system's crucial role in improving the precision of therapeutic interventions, particularly cancer treatment. This review underlines the importance of ongoing research in optimizing these systems for clinical applications.

Sam et al. [57] reviewed the versatile applications of temperature-responsive nanogels in drug delivery. These nanogels react to temperature through physical changes and outline the new trend in drug release mechanisms. Such potentialities make these candidates very potential for stimuli-responsive drug delivery that may specifically target a very nuanced field, such as cancer therapy, where very controlled drug release would finally yield significantly enhanced therapeutic outcomes. This mini-review emphasizes, on the one hand, some current advances in the niche but also opens a discussion regarding the possible integration of temperature-responsive nanogels into myriads of drug-delivery systems, pointing out their significance within modern pharmacotherapy.

The temperature-responsive NPs developed by Prawatborisut et al. [58] are also the first of their kind and can contribute to the selective binding of apolipoproteins from human plasma. The results showed how specific the target was and how temperature-responsive NPs might be able to target and bind a particular protein. This could open the door to targeted delivery and therapy. Lastly, Yuba's [59] review paper gave an in-depth look at the planning and creation of functional liposomes for new biomedical uses involving materials that respond to stimuli. It highlighted several areas where further research could be done in this area. The study will further highlight that temperature-sensitive liposomes, or NPs, are considered very flexible drug carriers as they can be modified according to the drug's therapeutic needs. In conclusion, these studies show that temperature-sensitive nanocarriers for drug delivery are a dynamic and ever-changing landscape that gives us new and exciting ways to think about this technology's role in the future of personalized medicine and targeted cancer therapies.

Light-responsive materials designed for precision medicine offer a new revolution, allowing for the non-invasive, on-demand, site-specific delivery of therapeutic agents with unprecedented spatial and temporal control. Smart materials activate the system from the outside, allowing precise location of drug-delivery nanocarriers where the medicine is needed and, in this case, sparing side effects associated with general, systemic drug administration. Introduced first by Liu et al. [60], singlet-oxygen-responsive linkers equipped with nanoscale coordination polymers are pioneering. This design lets light-controlled drug release happen for the first time when light is used with cancer therapy and achieves precise control of the therapeutic output.

Building on this earlier work, Xiang et al. [61] developed UV-responsive diblock copolymer-coated upconversion NPs. Thus, the upconversion NPs showed potential for photo-controlled drug delivery and gave a perfect example of the adaptation and efficiency of light-responsive materials in drug release. This opens the door to using the Hao Ma group's red-light-sensitive Donor–Acceptor Stenhouse Adduct (DASA) polymer with high water stability to control drug release in aqueous environments [62]. This horizon of light-responsive polymers has great potential for more effective and safer drug-delivery systems, whether they are established or not.

Rivero-Buceta et al. [63], on the other hand, utilized the “ship in a bottle” principle in one of their inventions and manufactured camptothecin porous materials to achieve a controlled and photoinduced release of light. Using light as a stimulus to release chemotherapeutic agents, they describe a work that emphasizes current strategies in innovative drug-delivery systems. The purpose of this work was to improve the precision of the application of these agents in the treatment of cancer. The first of these pioneering works underlined the great potential of light-responsive materials for developing controlled drug-delivery systems. Such materials will open new avenues in the progress of personalized medicine, wherein treatments could be applied only in the place of interest with extremely high precision for better efficacy and safety.

Smart, temperature, and light-stimuli-responsive nanocarriers represent considerable breakthroughs in research for drug delivery. Using temperature-sensitive liposomes, NPs, and light-responsive materials together could help fine-tune drug release rates for a therapeutic intervention that is both safe and effective. In this regard, further research and development are, therefore, required to be carried out in the field to actualize the optimum potential of such highly innovative, sophisticated drug-delivery systems in clinical practice.

2.2.3 Magnetic and Ultrasound-Responsive Systems

Integrating magnetic and ultrasound-responsive mechanisms into nanocarriers signifies a pivotal advancement in targeted therapy. These sophisticated systems introduce innovative solutions for drug delivery, utilizing non-invasive control methods to ensure precise delivery of therapeutic agents to the intended site, thereby minimizing side effects. The ability to finely tune drug release through external magnetic fields or acoustic waves significantly enhances therapeutic efficiency while prioritizing patient safety.

Wang et al. [64] highlighted the potential of combining magnetic targeting with focused ultrasound for intracerebral hemorrhage therapy. This dual approach enables concentrated therapeutic efforts in the affected area, improving treatment outcomes. Pawar et al. [34] emphasized integrating magnetic and ultrasound-responsive elements in nanocarrier designs for controlled drug release, showcasing their ability to be manipulated precisely in response to specific external stimuli.

In breast cancer treatment, Edwards et al. [65] explored ultrasound-responsive nanocarriers, suggesting a paradigm shift in treatment approaches. These carriers offer a way to reduce the side effects associated with traditional treatments while increasing the concentration of therapeutic agents directly at the tumor site. Zhu et al. [66] reviewed recent developments in magnetic nanocarriers for tumor therapy, focusing on the applications and specificity of magnetic iron oxide NPs in targeted drug delivery. This review underlines the significant role of magnetic nanoparticles (MNPs) in enhancing the specificity and efficiency of cancer treatments.

Additionally, Chapla et al. [67] discussed the potential of microbubble-NP complexes for ultrasound-enhanced drug delivery. This combination leverages the strengths of both microbubbles and NPs, offering a novel strategy for enhancing drug delivery across biological barriers and showcasing the capabilities of ultrasound-responsive systems in medical applications.

These studies highlight the advancements and promise of magnetic and ultrasound-responsive nanocarriers in nanomedicine. By offering precise, controlled drug-delivery mechanisms, these systems pave the way for more effective and less invasive therapeutic interventions, marking a new era in targeted therapy and personalized medicine.

MNPs stand as a cornerstone in advancing nanomedicine, especially for targeted drug delivery and controlled release. Their unique responsiveness to external magnetic fields allows for precise localization of drug delivery, significantly enhancing therapeutic efficacy and reducing systemic toxicity. This strategic drug-delivery approach overcomes challenges like nonspecific distribution and its resultant side effects. Liu et al. [68] introduced Cu₉S₅@mSiO₂@Fe₃O₄-PEG nanocarriers capable of facilitating magnetic resonance imaging alongside chemo/photothermal therapy. With the ability to use near-infrared photothermal therapy, this multifunctional platform shows how MNPs can be used for both the diagnosis and treatment of cancer.

Ghorbani et al. [69] designed ternary-responsive nanocarriers for targeted delivery of thiol-containing anticancer drugs, illustrating the potential of MNPs to improve therapeutic efficiency and patient-friendly cancer therapies. As emphasized by da Silva Oliveira and de Oliveira Conceição [70], magnetic drug-carrying NPs play a key role in treating cancer. They allow therapeutic agents to be magnetically targeted directly to tumor sites, which improves treatment outcomes.

Burmistrov et al. [71] explored the magneto-mechanical actuation capabilities of single-domain MNPs for controlled drug release, showcasing the flexibility of MNPs in creating responsive drug-delivery systems. Stanicki et al. [72] provided an update on the applications and characteristics of magnetic iron oxide NPs in drug-delivery systems, highlighting the evolving landscape of MNP-based therapeutics. Haghighattalab et al. [73] created a smart system for sustained drug release using a silk fibroin hydrogel reinforced with MNPs. This system shows how biocompatibility and responsiveness can be combined to create new drug-delivery solutions. Patel et al. [74] discussed the broad applications of MNPs in addressing therapeutic challenges, emphasizing their essential role in advancing targeted therapies.

These studies collectively underline the transformative impact of MNPs in drug delivery, offering new solutions that promise to revolutionize targeted therapy and controlled release mechanisms. Continuous research and development in this area are opening new doors, advancing the frontier of personalized and precision medicine.

Ultrasound-responsive nanocarriers represent a cutting-edge technology in drug delivery, using acoustic waves to control drug release. This method enhances tissue penetration and allows for minimally invasive therapeutic applications. Nanocarriers responsive to ultrasound are remarkably versatile, marking a significant advancement in nanomedicine by providing new solutions to improve therapeutic efficacy and patient compliance. Chapla et al. [67] highlighted the utility of microbubble-NP complexes in ultrasound-enhanced cargo delivery. Their research outlines a promising strategy for targeted tissue penetration and therapeutic enhancement. It utilizes the unique interactions between ultrasound waves and microbubbles for precise drug release and enhanced drug delivery to specific tissues.

Mukhopadhyay et al. [75] focused on the potential of ultrasound-responsive liposomes and polymeric micelles in cancer therapy. Their findings emphasized the selective nature of these carriers, significantly reducing toxicity and enabling controllable drug release patterns. This selectivity positions ultrasound-responsive nanocarriers as preferred vehicles for delivering anticancer drugs, offering a hopeful route to minimize side effects and improve therapeutic efficiency in cancer treatment. Tagai et al. [76] investigated drug-delivery systems with high responsiveness to ultrasound using metal NP-encapsulated liposomes. This approach aimed to enhance the response of liposomal carriers to ultrasound waves, thereby improving the efficacy of targeted drug delivery and controlled release mechanisms.

Additionally, the fabrication and evaluation of ultrasound-responsive emulsions loaded with paclitaxel for targeted chemotherapy were examined. This innovative strategy seeks to utilize ultrasound to trigger the release of chemotherapeutic agents directly at the tumor site, presenting a promising solution for enhancing cancer treatment efficacy while minimizing adverse effects. Ayana et al. [77] explored the development of ultrasound-responsive nanocarriers specifically for breast cancer chemotherapy. Their work demonstrates the potential of ultrasound-responsive nanocarriers to significantly improve the delivery and efficacy of chemotherapy agents in treating breast cancer, showcasing a compelling example of the therapeutic applications of these advanced nanocarriers.

These studies collectively highlight the transformative potential of ultrasound-responsive nanocarriers for targeted drug delivery and controlled release. As research in this field continues to evolve, the integration of ultrasound-responsive mechanisms in nanocarrier design is poised to revolutionize therapeutic interventions, offering more efficient, targeted, and patient-friendly treatment options.

2.3 CHALLENGES AND FUTURE DIRECTIONS

The development and use of nanocarriers for drug delivery and diagnostics have recently advanced thanks to advancements in nanotechnology and a better understanding of biological processes. However, transitioning these innovations from the laboratory to clinical application is a complex journey fraught with challenges that must be diligently addressed to fully realize their transformative potential. Li et al. [78] discussed the challenges and opportunities presented by nanomaterials in mRNA-based therapeutics, highlighting the need to solve issues related to stability, delivery efficiency, and immune response modulation. This underscores the importance of overcoming these obstacles to maximize the therapeutic potential of nanocarriers in gene therapy.

Cai et al. [79] focused on the physicochemical properties, pharmacokinetics, toxicology, and applications of nanocarriers, emphasizing the need for a perfect balance to utilize them clinically. This study points to the significance of understanding nanocarriers' physicochemical and biological interactions to enhance therapeutic effectiveness and reduce toxicity. Ahmad et al. [80] extended the analysis into precision nanotoxicology in drug development, addressing customized multifunctional nanocarriers' safety and toxicity implications. This highlights the critical need for safety assessments in developing nanocarriers for clinical use.

Dri et al. [81] provided insights into the state of clinical trials with nanomedicines and nanocarriers, including regulatory requirements and physicochemical critical quality attributes. Their analysis of the regulatory landscape offers perspectives on the conditions needed for the successful transition of nanocarriers from the bench to the bedside, emphasizing the necessity of adhering to regulatory standards.

These contributions underline the complexity of translating nanocarrier innovations into clinical applications. A multidisciplinary approach involving nanotechnology, biology, pharmacology, toxicology, and regulatory science is essential to overcome the identified difficulties. The ongoing contribution of these advancements promises significant patient benefits across various therapeutic areas as the field of nanomedicine continues to proliferate.

The field of nanocarrier design faces several critical challenges that need to be addressed for a successful transition from laboratory innovation to clinical applications. These challenges encompass various aspects of nanocarrier development, including reproducibility, scalability, regulatory issues, and the safety and biocompatibility of these systems.

Fernandes et al. [82] tackled the scale-up challenges associated with NP manufacturing processes. The difficulty of consistently producing nanocarriers with uniform size, shape, and functional properties presents a significant hurdle. This inconsistency impacts the scalability of nanocarrier manufacturing processes, which are essential for clinical applications. Patel et al. [83] discussed the complexities involved in the preformulation of nanoformulations and the intricacies in their selection, design, and preparation. The complexity of nanocarriers introduces substantial regulatory challenges, necessitating thorough characterization and standardization to ensure their safety, efficacy, and quality. The regulatory frameworks need to evolve to adequately address the unique challenges posed by nanomedicine.

Patel et al. [84] explored the challenges and feasibilities of scaling up nanomedicine technologies. Despite the promising potential of nanocarriers for targeted delivery and minimizing side effects, concerns regarding their long-term safety and biocompatibility remain. Understanding the interactions of nanocarriers with biological systems is crucial to mitigating their potential immunogenicity and toxicity. Cury et al. [85] highlighted the importance of designing nanocarriers to enhance drug absorption through the gastrointestinal tract, addressing one of the many challenges requiring innovative nanocarrier applications.

These works collectively emphasize the complex landscape of nanocarrier design and development. Addressing these challenges necessitates a multidisciplinary approach that includes insights from material science, pharmacology, toxicology, and regulatory science. As the field progresses, ongoing research and collaboration among scientists, clinicians, and regulatory bodies will be crucial in overcoming these obstacles, paving the way for the successful clinical translation of nanocarriers and realizing their full potential in enhancing patient outcomes.

The future of nanocarrier design is bright, with several promising directions poised to overcome current limitations and significantly enhance therapeutic outcomes. Among the notable advancements are the development of multi-stimuli-responsive systems, the integration of artificial intelligence (AI) and machine learning (ML) into nanocarrier design and optimization, and the move toward personalized nanomedicine. Modern nanogels are being engineered to respond to multiple stimuli, such as pH, temperature, or magnetic fields, offering enhanced drug release and targeting control. For example, recent research has demonstrated the effectiveness of multi-stimuli-responsive and cell membrane-camouflaged Aggregation-Induced Emission (AIE) nanogels for chemo-photothermal synergistic therapy of tumors, highlighting their adaptability and responsiveness in complex biological environments [86].

Applying AI and ML technologies in nanocarrier design opens new frontiers, enabling predictions about nanocarrier behavior, optimizing formulations, and personalizing therapies based on patient-specific data. This approach has already shown promise, particularly in magnetic hyperthermia research, where AI applications have improved our understanding and use of nanocarriers for targeted therapies [87]. Tailoring nanocarrier systems to individual patient profiles and disease states is an emerging trend that could transform treatment strategies. By maximizing efficacy and minimizing adverse effects, personalized nanomedicine focuses on patient-centered approaches, offering solutions based on individual health data [88].

Stimuli-responsive multifunctional nanomedicine presents a novel strategy for enhancing chemotherapy, particularly for complex conditions like glioblastoma. By improving blood-to-brain trafficking and tumor targeting, these approaches highlight the importance of multifunctional and responsive nanocarriers in addressing treatment complications [89].

These advancements in nanomedicine, including AI and ML, multi-stimuli-responsive systems, and personalized approaches, are expected to play pivotal roles in overcoming the current challenges of nanocarrier design and application. These future directions signify a shift towards more sophisticated, efficient, and patient-specific nanotherapeutic strategies, potentially revolutionizing drug delivery and diagnostics.

Advancing in nanomedicine necessitates a multifaceted approach to surmount the inherent challenges of transitioning nanocarrier technologies from lab to clinical settings. This journey is a collaborative effort that melds diverse expertise, state-of-the-art manufacturing, and sophisticated computational models to unlock nanocarriers' full therapeutic potential, making them fundamental to advanced healthcare solutions. Nanomedicine's complexity demands a multidisciplinary team incorporating materials science, biology, pharmacology, and regulatory science. Each discipline contributes unique insights crucial for tackling nanocarrier design, safety assessment, and therapeutic application challenges. Fostered by open communication and shared objectives, interdisciplinary teams can spur innovation, from novel nanocarrier designs to strategies for mitigating risks and boosting clinical efficacy.

Realizing nanomedicine's promise requires advancements in manufacturing technologies for scalable, reproducible, and high-quality nanocarrier production. Emerging nanofabrication techniques, along with continuous process optimization, are essential for meeting clinical demands. Developing standardized protocols and quality controls will aid regulatory approval, transitioning nanomedicine from experimental therapies to global treatments. Integrating AI and ML opens unparalleled opportunities for enhancing nanocarrier design and optimization. These computational tools can predict nanocarrier behavior within biological systems, optimize formulations across numerous parameters, and tailor therapy regimens to individual patient data, accelerating nanomedicine development and introducing a new paradigm in precision medicine.

The future of nanomedicine lies in personalized therapeutic solutions. Tailoring nanocarrier systems to individual patient profiles and disease states promises to revolutionize treatment strategies. This personalization demands a deep understanding of disease molecular mechanisms and integrating

patient-specific data into the nanocarrier design process, resulting in more effective, safer, and patient-friendly therapies. While the path forward for nanomedicine is challenging, it is filled with opportunities for breakthrough advancements. We can unlock nanocarriers' potential by fostering interdisciplinary collaboration, pushing manufacturing technologies forward, harnessing computational insights, and moving toward personalized medicine. This concerted effort will advance nanomedicine and herald a new healthcare era where nanotechnology significantly improves patient outcomes and life quality.

2.4 CONCLUSIONS

The exploration and utilization of nanocarriers for delivering nucleic acids and proteins represent a significant leap in medical therapeutics, unveiling novel strategies to circumvent traditional drug-delivery challenges. Through meticulous design and innovative breakthroughs, nanocarriers have demonstrated exceptional potential for enhancing nucleic acid and protein delivery systems' effectiveness. Their capability to precisely target disease sites, minimize off-target effects, and systemic toxicity marks a considerable advancement in therapeutic modalities.

Nanocarriers' ability to be engineered to target specific cells or tissues using biological pathways is a major breakthrough that improves the therapeutic index of the drugs they carry and ensures direct delivery to the right cells or tissues. Additionally, integrating stimuli-responsive materials into nanocarrier designs enables controlled therapeutic agent release in response to specific environmental triggers at precisely needed locations. Moreover, encapsulating therapeutic agents within nanocarriers significantly boosts their stability and solubility, facilitating the delivery of otherwise challenging compounds.

Despite these advancements, challenges such as reproducibility, scalability, and regulatory compliance remain. Addressing these challenges necessitates ongoing research and innovation in various areas. Developing multi-stimuli-responsive nanocarriers promises finer control over drug release and targeting, adapting dynamically within complex biological environments. Integrating AI and ML into nanocarrier design could revolutionize this field, enabling behavior prediction, formulation optimization, and personalized therapies based on individual patient data. Furthermore, personalizing nanomedicine to align with individual patient profiles aims to maximize efficacy and minimize adverse effects, heralding a shift toward patient-centric therapeutic approaches.

Overcoming current obstacles and releasing nanocarriers' full therapeutic potential are crucial as the field develops as a result of multidisciplinary research, technological advancements, and clinical insights. Although the path from lab to clinical application is fraught with challenges, the transformative potential for medical therapeutics is unparalleled. It heralds a new era of precision medicine and targeted therapies, potentially revolutionizing disease treatment methodologies.

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